TECHNICAL OVERVIEW

Suar

How the app works — presence, GPS, and a Bluetooth-only mesh, with no internet or cellular connection required.

> Suar lets people find each other and reach a marked "assembly point" during a disaster, when cell towers and Wi-Fi are down. Two or more phones exchange presence and GPS location purely over Bluetooth Low Energy advertising — no pairing, no internet, no server.

1 What Suar does

Every phone running Suar does two things at once: it **broadcasts** its own presence (a short-range radio "I am here, at this GPS position") and it **listens** for the same broadcast from everyone nearby. There's no login, no pairing, no data connection of any kind — the entire app works over Bluetooth Low Energy (BLE) advertising packets, the same mechanism used by beacons and fitness trackers, which any phone can send and receive without a network.

A phone that's out of BLE range of you can still show up on your map: nearby phones **relay** each other's broadcasts, decrementing a hop counter each time, forming a short-range mesh. One phone can also mark itself as a fixed **assembly point** — a "come here, this is safe" beacon that everyone else's radar can home in on.

No infrastructure

GPS is satellite-based and BLE is peer-to-peer — both keep working with zero signal bars, zero Wi-Fi, and zero cell towers. That's the entire premise of the app.

Honest positioning

A dot on the map is only ever a real GPS delta between two phones. When the two devices' combined GPS error is too large to trust a bearing, the beacon is flagged approximate or moved to a "direction unknown" list — never silently guessed.

Multi-hop mesh

Every relayed frame is itself eligible for further relay (up to 3 hops), so presence can travel through a chain of phones even when the two endpoints never hear each other directly.

Accessible by default

Voice guidance, haptic proximity feedback, and the visual map all run simultaneously. Settings only let a user turn a channel *off* — everything is on from first launch.

2 End-to-end flow

After permissions and Bluetooth are confirmed, the app runs several loops concurrently: it watches its own GPS position and compass heading, rebuilds and broadcasts its own presence on a timer, and continuously scans for and relays everyone else's. All of it feeds one shared beacon store, which drives the on-screen map.

Start / trigger   Process step   Decision   Stored state   Screen / output

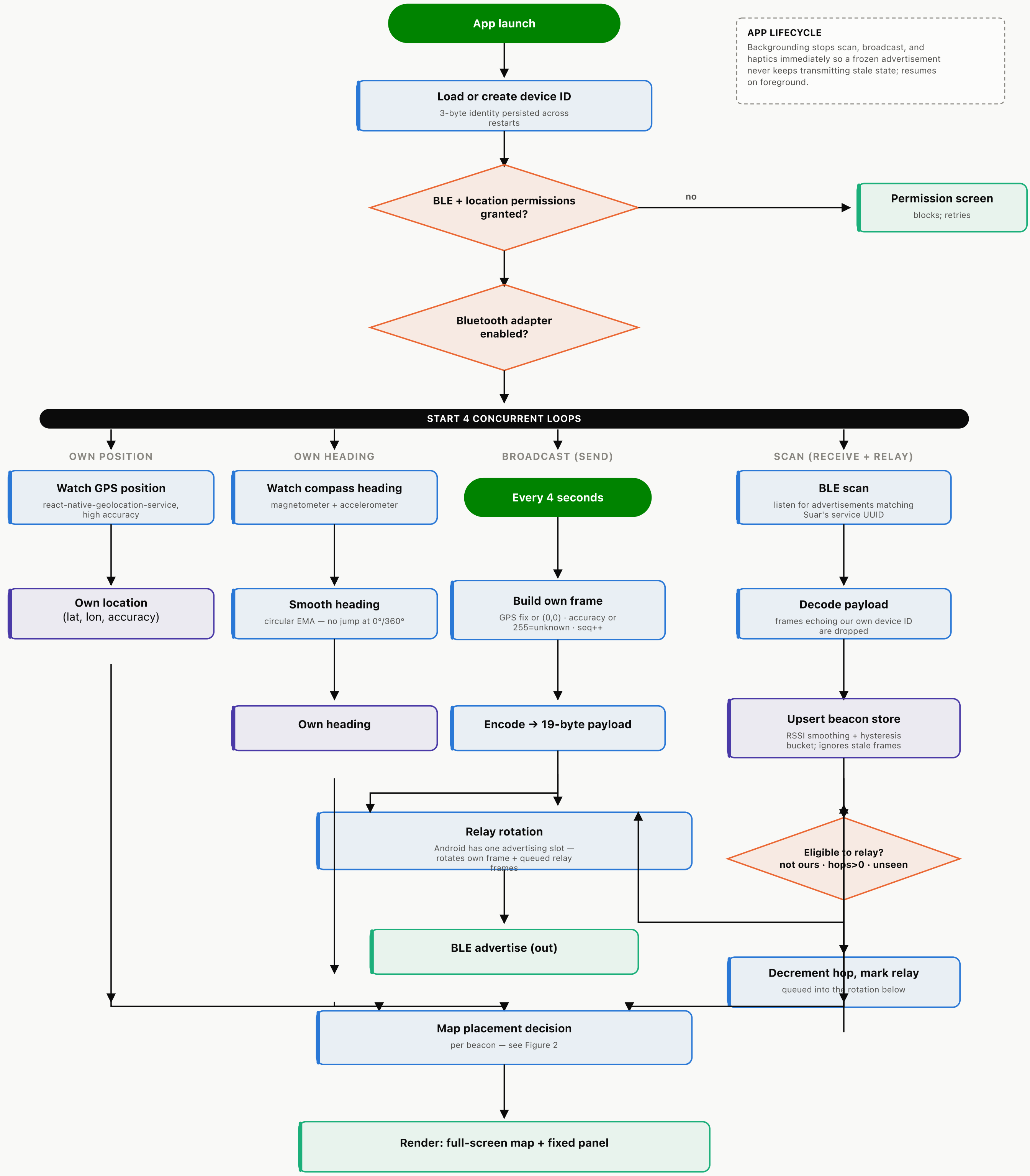

Figure 1. Startup gate, the four concurrent loops, and where they converge on the beacon store and the map. A beacon that fails the relay check (it's ours, already relayed once, or already seen) is simply left out of the rotation — nothing further happens to it.

3 What actually goes over the air

Every broadcast is a fixed **19-byte** payload, packed to fit inside the ~20-byte manufacturer-data budget a single legacy BLE advertisement can carry. No text, no JSON — just packed integers, decoded identically on every receiving phone.

BYTES	FIELD	CONTENTS
0	Flags	Beacon type (Person/Assembly) · is-relay flag · protocol version · hops remaining (4 bits, 0–15)
1–3	Device ID	3-byte identity, generated once and persisted — the same phone always shows as the same dot, even after a restart
4–7	Latitude	Signed 32-bit integer, degrees × 10,000,000 — sub-metre precision without a decimal point
8–11	Longitude	Signed 32-bit integer, same fixed-point encoding
12–15	Timestamp	Unix seconds — lets a receiver ignore an older frame that arrives after a newer one
16–17	Sequence	Per-device counter, wraps at 65,536 — de-duplicates relayed frames
18	Accuracy	GPS accuracy in metres, 0–254 (255 = unknown) — lets a *receiver* judge how much to trust the sender's position

> Before a GPS fix arrives, a phone broadcasts latitude/longitude as `(0, 0)` and accuracy as `255` — an explicit "no fix yet" placeholder every receiver checks for, rather than a missing field that could be misread as the equator.

4 Deciding what a dot is allowed to claim

The most important rule in the app: a dot's position on the map is a claim about where someone *is*, so it is never drawn from a guess. Every beacon is placed by asking two separate questions, each with its own honesty threshold.

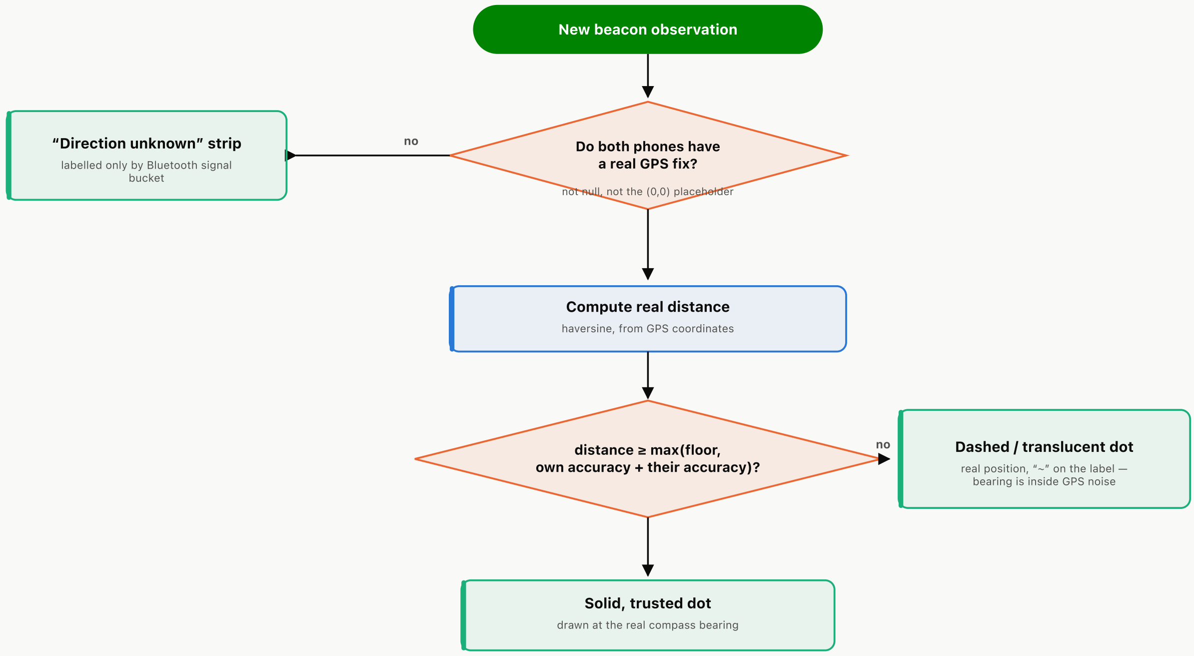

Figure 2. Every discovered beacon runs through this on every update. The bearing gate widens with either phone's GPS error, so a device with a coarse fix needs proportionally more separation before it's drawn with confidence.

On the map

Trusted beacons draw as a solid dot. Beacons inside the bearing gate still draw — same real GPS delta — but dashed/translucent with a "~" on the distance label, because a real vector between two very close phones is still dominated by GPS noise.

Off the map

A beacon with no GPS fix at all (either phone still on the `(0,0)` placeholder) has no position to plot, so it lives in the "Direction unknown" strip below the map, labelled only by coarse Bluetooth signal strength (Very near / Near / Medium / Far).

5 What's on screen

The map

Fills the screen edge-to-edge. "You" sit fixed at the centre with an arrow; in heading-up mode the whole world — grid, range rings, every dot — rotates under that arrow using the phone's compass, so "up" always means "the way you're facing." Zoom steps through 50/150/300 m, each with grid lines and two labelled range rings for scale. A beacon further than the current zoom clamps to the edge as an outward-pointing chevron rather than disappearing.

The fixed panel

A rounded, drag-handle-styled panel pinned to the bottom of the screen — it looks like a bottom sheet but is deliberately static, not draggable. It holds the "direction unknown" strip and the full scrollable list of every beacon with distance, signal bucket, and an explicit "approximate position" note where it applies.

Settings

A real modal sheet (distinct from the fixed panel above): toggle this phone as an assembly point, and switch off voice guidance, haptics, or enable a battery-saving disaster mode. Every channel is on by default — settings only ever turn one off.

Voice & haptics

Speaks distance and a hot/cold trend to the focused (or nearest) beacon on an interval — but never a bearing, since a spoken "turn right" needs more compass confidence than a visual rotation does. Vibration follows the same target through four proximity buckets.